

A rare mutation in the H3N2 influenza virus leads to antigenic drift

Bogdanova Liliia^{1,3} and Polovinkin Mikhail^{2,3}

¹Sechenov University

²Department of Statistical Physics, St. Petersburg State University

³Bioinformatics institute

Abstract

An essential cause of disease death in the world still comes from influenza. New mutations that appear in the influenza genome are partially responsible for the persistent threat of infection by this virus. The hemagglutinin (HA) glycoprotein is one of the gene products that plays an essential role in influenza pathogenesis and is the target of vaccines. Over time, HA develops new mutations that can reduce vaccine effect ([Cushing et al. 2015](#)). In this work we analyze sequencing data from a person who was infected with the human influenza virus H3N2. Afterwards, we identified the mutation that resulted in an amino acid substitution in a protein known as the epitope, which is recognized by antibodies. Therefore, SNP was found at position 307, epitope D ([Muñoz and Deem 2005](#)). Proline was switched out for serine as a result of the mutation (P103S), which could have led to breakthrough infection.

Keywords: Influenza virus; Hemagglutinin; Epitope D

Introduction

Influenza viruses are members of the family Orthomyxoviridae ([Blümel et al. 2008](#)). Based on differences in the nucleoprotein antigen, influenza viruses are classified as types A, B, and C. New strains are created as a result of genetic variation in the hemagglutinin and neuraminidase antigens of types A and B ([editor S. Baron 1996](#)).

The main envelope glycoprotein of A and B viruses is hemagglutinin (HA), a protein that is similar to HA. When complete proteins are formed, HA is broken into the signal peptide (approximately 20 amino acids in influenza A viruses), protein HA1 (about 320 amino acids), and protein HA2 (about 220 amino acids). HA1 is a protein that binds to receptors and is the main target of antibodies that confer protective immunity to influenza viruses, whereas HA2 is an envelope anchor protein that facilitates the fusing of the envelope with the cellular endosomal membrane ([Suzuki and Nei 2002](#)). Influenza virus hemagglutinin (HA) includes antigenic sites recognized by the host immune system, cleavage sites broken down by host proteases, fusion peptides that facilitate membrane fusion and receptor binding sites connecting to receptors on the target cell. A change in an amino acid at these locations may have an impact on the likelihood of viral infection and transmission both within and between hosts ([Suzuki 2012](#)).

In influenza A, different combinations of the hemagglutinin and neuraminidase antigens are present. These antigens go through antigenic drifts and shifts, resulting in antigenic variation, needing various vaccine strain types. Antigenic drifts are genetic modifications, which gradually alter both HA's antigenic composition and result in the emergence of new variant strains ([Kalarikkal and Jaishankar 2022](#)). The influenza virus's poor fidelity RNA polymerase, which lacks

an error-proofing mechanism, is responsible for antigenic drift. Polymerase errors cause de novo mutations during each replication cycle, broadening the genetic diversity of the virus. As a result, mutations in the HA gene can arise and, depending on their location, may change the antigenic epitopes of HA ([Cushing et al. 2015](#)). By encouraging the creation of antibodies specific to the illness, the influenza vaccination provides protection against the influenza virus. When antibodies against HA are present, viruses may efficiently aggregate on the cell surface and less virus is discharged from infected cells ([Kalarikkal and Jaishankar 2022](#)).

The evolution of next generation sequencing (NGS), also known as deep sequencing technology, provides a potent tool for researching the inherent heterogeneity of nucleic acids. It marks a significant advancement over cloning and clone sequencing in the investigation of the quasispecies nature of RNA viruses. Because their polymerase is inefficient and causes point mutations during replication, RNA viruses persist as heterogeneous populations of closely related genetic variations and are referred to as quasispecies ([Barbezange et al. 2018](#)).

Models and methods

A code describing our work can be found [here](#). The raw sequencing data of flu in FASTA format was loaded from Sequence Read Archive (SRA) ([Raw data link](#)), it is similar to A/Hong Kong/4801/2014 (H3N2) strain. Data was examined with the fastQC tools ([FastQC](#)). The control sample sequencing data was provided by SRA ([Control1](#), [Control2](#), [Control3](#)). The sequencing data was obtained via Illumina technology ([Illumina](#)). The influenza hemagglutinin gene reference sequence was obtained from National Center for Biotechnology Information (NCBI) database at [Reference](#). The sequencing data (both sequencing and control) was indexed and aligned to the reference em-

playing Burrows-Wheeler Alignment (BWA) package (Li and Durbin 2009). The indexing and alignment were performed with default settings. The Samtools software (Danecek et al. 2021) was used for indexing of alignment files and obtaining mpileup files, the depth limit in mpileup was removed (parameter -d 0). Common and rare variants in sequencing data were found via VarScan software (Koboldt et al. 2009, 2012) with variant frequency threshold 0.95 (parameter -min-var-freq 0.95) and 0.001 (parameter -min-var-freq 0.001) respectively.

The protein structure was obtained with swiss model software (Waterhouse 2018). This code takes an amino acid sequence as an input and provides the predicted structure in PDB format. The VMD tool (Humphrey et al. 1996; Eargle et al. 2006) was employed for visualisation of the substitution in protein.

Statistical analysis

The variant calling with low variant frequency threshold determines a plenty of SNPs, but not all of them are actually real mutations in sequence. There are also mistakes in amplification process and even Illumina sequencing misreads. So we need some method to filter these errors. The frequency approach was used. The main idea is to analyse the material without any mutations in order to obtain the variant frequency and standard deviation of sequencing and preprocessing errors. The average and standard deviation were calculated as

$$\langle f \rangle = \frac{1}{N} \sum_{i=1}^N f_i, \quad (1)$$

where f - frequency of variant, N - number of errors, f_i - frequency of error i ,

$$\sigma = \sqrt{\langle f_i^2 \rangle - \langle f \rangle^2}, \quad (2)$$

where σ - is a standard deviation, f_i - frequency of error i , $\langle \cdot \rangle$ is an averaging over all errors. Then one should understand if any of the variants in studying sequencing lie outside the three standard deviations from mean frequency of errors in control. Thus, these variants are emissions with respect to the error rate in control and appears to be the real mutations. The excel software was employed to calculate the statistics in our case.

The following approach is suggested to distinguish sequencing and preprocessing errors in three control samples. One can gather statistics over number of mistake occurrence in the current position, calculate how many times error in the current position occurs in control samples. The probability of two or more sequencing mistakes appearance in the same position is likely to be extremely low. Hence, the errors with occurrence rate of 1 are of sequencing nature, otherwise belong to amplification processes. In house program was used to obtain occurrence statistics. The realisation was the following: obtain a file with all SNP over control samples in VCF-like format, parse the position of every SNP and increase by 1 the array element with the number equal to position. The produced array contains the number of errors in the current position. This array can be analysed later.

Results

The raw sequencing data was examined in order to determine the quality of it. The quality of raw data was fine, thus, the trimming was not performed. The sequencing data alignment

Read	Number of reads	Mapped reads
Studied Sample	358265	358032
Control Sample 1	256586	256500
Control Sample 2	233327	233251
Control Sample 3	249964	249888

Table 1 The statistics over the initial and mapped number of reads in data and control files.

was processed. An initial number of reads and a number of mapped reads are represented in table 1. The control sequencing files were examined, the average frequencies and their standard deviations were calculated over three files in separate. The general frequency and it's standard deviation was also found in order to realise the idea described in Statistical analysis. The frequencies were the following (see table 2). Thus, the resulting frequency threshold for real mutation is 0.46.

Alignment	Average frequency of error	Standard deviation
Control 1	0.25	0.08
Control 2	0.24	0.05
Control 3	0.26	0.07
In general	0.25	0.07

Table 2 The average frequencies and standard deviations of errors in control files.

The variant calling derived 21 mutations which are represented in table 4. Only 7 of them are beyond the variant frequency threshold of errors and considered to be real. One can see analysis in Discussion section.

As for the errors' nature, the approach of distinguishing sequencing and preprocessing errors in control data was realised. Errors position distribution was obtained via in house program and is represented at figure 1.

There are 33 errors of rate 3, 19 of rate 2 and 33 of rate 1, the number of positions without errors is 1565, 1650 positions in general.

Discussion

Seven mutations were found by our analysis that weren't the result of sequencing or preprocessing. Our analysis revealed seven mutations that were not the result of poor sequencing or pretreatment. Five of them were found in more than 99% of viral RNA molecules, while the two others were identified as emissions using a frequency-based approach. Only one mutation, in 307 position is not synonymous, as a result of this mutation proline has been replaced by serine (P103S). This mutation is found in epitope D, which is part of the genome that codes for the HA1 protein, see figures 2 and 3. An antigen's epitope is the region of the antigen that binds to a specific antigen receptor on a B cell's surface. Only when the structures

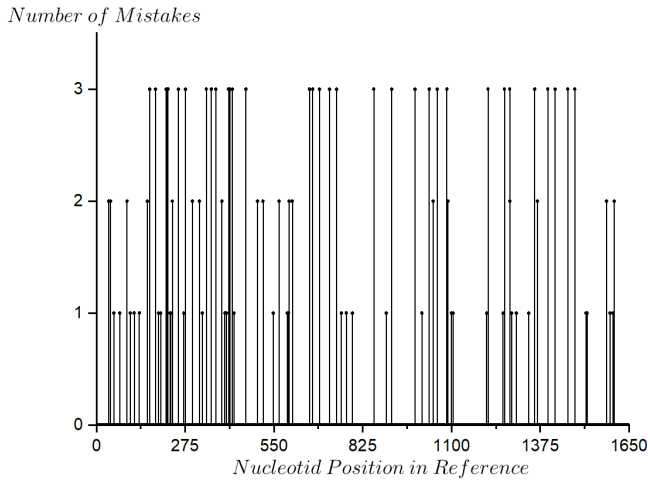


Figure 1 The number of mistakes in nucleotides' position dependence on the position. The horizontal axis is the position in reference and the vertical axis is the number of mistakes in it.

of the receptor and epitope are compatible does binding between them take place. The activation of B-cell production of antibodies depends on the epitope and receptor fitting together (Britannica 2019). The antigen receptor could become unable to identify the epitope structure if the structure of the epitope changes. Our mutation could lead to conformational and/or affinity changes, which significantly reduce the affinity of the antigen-binding site to epitope D. Variations caused by point mutations in the residues in the epitope areas can significantly affect the immune response (Muñoz and Deem 2005).

The current change in the structure of the protein is the first draft estimation and is desirable to be verified by experimental (x-ray diffraction analysis) or modeling (molecular dynamics) methods in further studies.

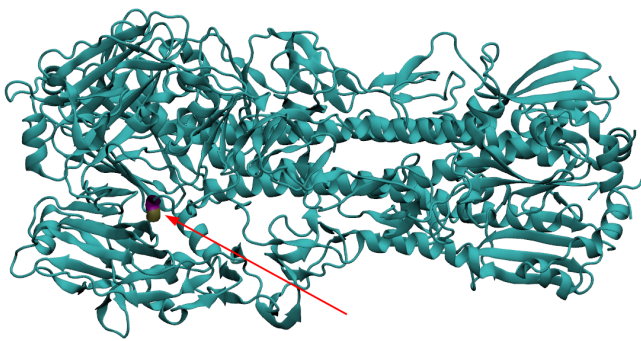


Figure 2 The swiss model predicted structure of reference HA protein. The red arrow indicates a place of substitute.

Based on the information presented above, we may assume that the viral population underwent mutation and evolution while replicating in human cells that had not got the vaccine. This is thought to have caused "antigen drift" and, as a result, the virus was not detectable by antibodies produced following vaccination.

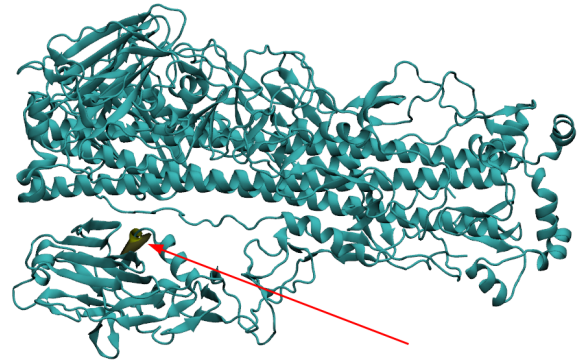


Figure 3 The swiss model predicted structure of HA protein after mutation. The red arrow indicates a place of substitute.

The figure 1 allows us to examine which of the errors have sequencing and amplification process nature. In suggested approach, the result claims there were more errors of amplification process then one of sequencing origin, 52 and 33 respectively.

There are different ways to avoid errors in sequencing or reduce an amount of them. By combining overlapping paired-end reads, removing reads with low-quality base calls, and, in the case of amplicon reads, clustering, quality filtering can minimize errors (Laehnemann et al. 2016). The another common approach of filtering of the errors is the following. From sequencing data k-mer coverage histogram is built. The true k-mer coverage distribution obeys the Poisson distribution. The strong perturbations of this distribution are caused by the sequencing mistakes. The threshold parameter can be chosen to reject the part of histogram with perturbations. The filtered k-mer distribution allows one to chose which reads should be implemented into true alignment. This method found in Musket corrector and BLESS 2 method. Musket is a powerful substitution-error-based corrector for Illumina sequencing data that uses a multistage K-mer spectrum method (Liu et al. 2013). BLESS 2 method is based on the use of the Bloom filter, as well as with the application of clipping reading, if the error cannot be corrected or corrected ambiguously (Heo et al. 2016).

An estimation of the actual coverage was made in the following way: one can calculate the number of bases mapped from sequencing to reference data and divide it by the genome size. For this purpose the coverage function of Samtools can be used. They should provide to it the reference sequence and aligned reads in sorted BAM-format. The output contains some properties of reads and reference data, the number of covered reference bases and it's percentage, average depth of coverage, and quality properties of sequencing and mapping. The coverage function was applied to our data, the result is represented in table 3.

```
samtools coverage --reference reference.fasta
Control1_sorted.bam Control2_sorted.bam Control3_sorted.bam
```

Reads	Covbases	%	AvgDepth	BaseQ	MapQ
740141	1665	100	65244.4	36.6	60

Table 3 The statistics of coverage over three control sequences.

The result correspond the naive estimation. The number of reads in all three sequences is equal to 739877, their average length is 150 (estimation from fastQC length distribution), the reference size is 1650, $\frac{739877 \times 150}{1650} \simeq 67261$.

References

- A. Cushing, A. Kamali, and M. Winters et al. Emergence of hemagglutinin mutations during the course of influenza infection. *Sci Rep* 5, 16178, 2015. doi: <https://doi.org/10.1038/srep16178>.
- Enrique T. Muñoz and Michael W. Deem. Epitope analysis for influenza vaccine design. *Vaccine*, pages 1144–1148, 2005. doi: <https://doi.org/10.1016/j.vaccine.2004.08.028>.
- J. Blümel, R. Burger, C. Drosten, A. Gröner A., L. Gürtler, M. Heiden, M. Hildebrandt, B. Jansen, H. Klammer, T. Montag-Lessing, R. Offergeld, G. Pauli, R. Seitz, U. Schlenkerich, V. Schottstedt, H. Willkommen, and C. H. von König. Influenza virus. transfusion medicine and hemotherapy : offizielles organ der deutschen gesellschaft für transfusionsmedizin und immunhamatologie. page 42–49, 2008. doi: <https://doi.org/10.1159/000111480>.
- editor S. Baron. Medical microbiology. 4th edition. galveston (tx): University of texas medical branch at galveston. page Chapter 58, 1996. doi: <https://www.ncbi.nlm.nih.gov/books/NBK8611/>.
- Y. Suzuki and M. Nei. Origin and evolution of influenza virus hemagglutinin genes. *Molecular Biology and Evolution*, Issue 4, 19:501–509, 2002. doi: <https://doi.org/10.1093/oxfordjournals.molbev.a004105>.
- N. Sriwilaijaroen Y. Suzuki. Molecular basis of the structure and function of h1 hemagglutinin of influenza virus. *Proceedings of the Japan Academy. Series B, Physical and biological sciences*, 88(6), page 226–249, 2012. doi: <https://doi.org/10.2183/pjab.88.226>.
- S. M. Kalarikkal and G.B. Jaishankar. Influenza vaccine. *Stat-Pearls [Internet]. Treasure Island (FL): StatPearls Publishing*, 2022. doi: <https://www.ncbi.nlm.nih.gov/books/NBK537197/>.
- C. Barbezange, L. Jones, H. Blanc, O. Isakov, G. Celniker, V. Enouf, N. Shomron, M. Vignuzzi, and S. van der Werf. Seasonal genetic drift of human influenza a virus quasispecies revealed by deep sequencing. *Frontiers in microbiology*, 9, 2596., 2018. doi: <https://doi.org/10.3389/fmicb.2018.02596>.
- H. Li and R. Durbin. Fast and accurate short read alignment with burrows-wheeler transform. *Bioinformatics*, 25(14):1754–1760, 2009. doi: [10.1093/bioinformatics/btp324](https://doi.org/10.1093/bioinformatics/btp324).
- P. Danecek, J.K. Bonfield, J. Liddle, J. Marshall, V. Ohan, M. O. Pollard, A. Whitwham, T. Keane, S.H. McCarthy, R. M. Davies, and H. Li. Twelve years of samtools and bcftools. *GigaScience*, 10(2), 2021. doi: [10.1093/gigascience/giab008](https://doi.org/10.1093/gigascience/giab008).
- D. Koboldt, K. Chen, T. Wylie, D. Larson, M. McLellan, E. Mardis, G. Weinstock, R. Wilson, and L. Ding. VarScan: Variant detection in massively parallel sequencing of individual and pooled samples. *Bioinformatics*, 25(17):2283–2285, 07 2009. doi: [10.1093/bioinformatics/btp373](https://doi.org/10.1093/bioinformatics/btp373).
- Daniel Koboldt, Qunyuan Zhang, David Larson, Dong Shen, Michael McLellan, Ling Lin, Christopher Miller, Elaine Mardis, Li Ding, and Richard Wilson. VarScan 2: Somatic mutation and copy number alteration discovery in cancer by exome sequencing. *Genome research*, 22:568–576, 2012. doi: [10.1101/gr.129684.111](https://doi.org/10.1101/gr.129684.111).
- Bertoni M. Bienert S. Studer G. Tauriello G. Gumienny R. Heer F.T. de Beer T.A.P. Rempfer C. Bordoli L. Lepore R. Schwede T. Waterhouse, A. Swiss-model: homology modelling of protein structures and complexes. *Nucleic Acids Res.*, 46:W296–W303, 2018.
- William Humphrey, Andrew Dalke, and Klaus Schulten. Vmd – visual molecular dynamics. *Journal of Molecular Graphics*, 14: 33–38, 1996.
- John Eargle, Dan Wright, and Zaida Luthey-Schulten. Multiple alignment of protein structures and sequences for vmd. *Bioinformatics*, 22(4):504–506, 2006.
- T. Editors of Encyclopaedia Britannica. Epitope analysis for influenza vaccine design. *Encyclopedia Britannica.*, 2019. doi: <https://www.britannica.com/science/epitope>.
- David Laehnemann, Arndt Borkhardt, and Alice Carolyn McHardy. Denoising dna deep sequencing data—high-throughput sequencing errors and their correction, briefings in bioinformatics. 1:154–179, 2016. doi: <https://doi.org/10.1093/bib/bbv029>.
- Yongchao Liu, Jan Schröder, and Bertil Schmidt. Musket: a multistage k-mer spectrum-based error corrector for illumina sequence data. *Bioinformatics*, 29:308–315, 2013. doi: <https://doi.org/10.1093/bioinformatics/bts690>.
- Y. Heo, A. Ramachandran, W. M. Hwu, and Chen D. Ma, J. Bless 2: accurate, memory-efficient and fast error correction method. *Bioinformatics (Oxford, England)*, 32:2369–2371, 2016. doi: <https://doi.org/10.1093/bioinformatics/btw146>.

Position	Nucleotide in Reference	Nucleotide in Sequence	Variant Frequency, %	Type	Amino Acid Substitution
72	A	G	99.96	Common	T → T
117	C	T	99.82	Common	A → A
254	A	G	0.17	Error	
276	A	G	0.17	Error	
307	C	T	0.94	Rare	P → S
340	T	C	0.17	Error	
389	T	C	0.22	Error	
691	A	G	0.17	Error	
722	A	G	0.2	Error	
744	A	G	0.17	Error	
774	T	C	99.96	Common	F → F
802	A	G	0.23	Error	
859	A	G	0.18	Error	
915	T	C	0.19	Error	
999	C	T	99.86	Common	G → G
1043	A	G	0.18	Error	
1086	A	G	0.21	Error	
1213	A	G	0.22	Error	
1260	A	C	99.94	Common	L → L
1280	T	C	0.18	Error	
1458	T	C	0.84	Rare	Y → Y

Table 4 The mutations obtained with VarScan and their analysis.